Lectures 10-11

Chapter 5: Continuous time Markov Chains

References

Oliver Ibe – Markov Processes for Stochastic Modeling (Chapter 5)

Sheldon Ross – Introduction to Probability Models (Chapter 6)

Introduction

Markov Chains

Poisson Processes and Pure Birth Process

Birth and Death Processes

Continuous Time Markov Chains

Markovian Property

For an SP, $s, t \geq 0$, non-negative integers i, j and $x(u)$ where $0 \leq u < s$

$$P[X(t+s) = j | X(s) = i, X(u) = x(u), 0 \leq u < s] = P[X(t+s) = j | X(s) = i]$$

Homogeneous probabilities

Holding Time

Probability Models for CTMC

1. States of the MC

2. Holding Time

3. Transition from the present state to any other states

Distribution of the Holding Time

Two possible scenarios:

1. Keeping track of the time already spent in state i

2. Forget the time already spent in state i

Alternate Definition of CTMC

An SP with

1. Holding time $\sim$ Exponential with rate v_i

2. Future state is independent of the holding time of the current state

3. Future state of the process depends only on the current state, and a transition probability exists

Ex 4.1.2 Consider a shoeshine shop consisting of 2 chairs–chair 1 and chair 2. A customer upon arrival goes initially to chair 1 and then to chair 2. The service times at two chairs are assumed to be independent rvs which are exponentially distributed with service rates μ_1 and μ_2. Suppose that potential customers arrive in accordance with a Poisson process having rate λ, and that a potential customers will only enter the system if both chairs are empty. Model the system state by CTMC.

Probabilities associated with a state

Transition Probability Function $P_{ij}(t)$

Probability Function $P_i(t)$

Transition probability p_{ij}

Chapman-Kolmogorov Equations

$$P_{ij}(s+t) = \sum_k P_{ik}(s)P_{kj}(t)$$

$$P_{ij}(s+t) = P[X(s+t) = j \mid X(0) = i]$$

$$= \sum_k P[X(s+t) = j, X(s) = k \mid X(0) = i]$$

Holding Time, Transition Rate and Transition Probabilities

T_i $\equiv$ Time that a MC stayed at state i before making a transition

T_i => dwell time/holding time

v_i $\equiv$ Rate of transition from state i

p_{ij} $\equiv$ Probability of transition from state i, after the expiration of the holding time

State Transition Rates

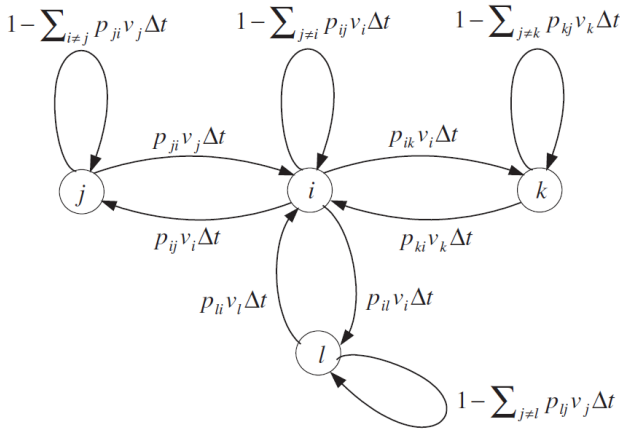
v_i => rate at which process leaves state i

$v_i p_{ij}$ => rate at which makes a transition to state j

Probability of a transition in a small time Δt, from i to j

Prob. of a transition from i to any state j

Prob. of no Transition from i



$$p_i(t+\Delta t) = p_i(t)\left\{1 - \sum_{j\neq i} p_{ij}v_i\Delta t\right\} + \sum_{j\neq i} p_j(t)p_{ji}v_j\Delta t$$

$$\lim_{\Delta t \to 0} \left\{ \frac{p_i(t + \Delta t) - p_i(t)}{\Delta t} \right\} = \frac{\mathrm{d}p_i(t)}{\mathrm{d}t} = -\, v_i p_i(t) \sum_{j \neq i} p_{ij} + \sum_{j \neq i} p_j(t) p_{ji} v_j$$

Birth and Death Processes

A birth and death process is a CTMC with states $\{0, 1, \cdots\}$ for which transitions from state n may go only to either state $n-1$ or state $n+1$.

Ex 4.2.1 Suppose there are n people in the system, then

i. new arrivals enter the system at an exponential rate λ_n, and

ii. people leave the system at an exponential rate μ_n.

Ex 4.2.2 Consider a birth and death process for which

$$\mu_n = 0, \qquad \text{for all } n \geq 0$$

$$\lambda_n = \lambda, \qquad \text{for all } n \geq 0$$

This is a process in which departures never occur, and the time between successive arrivals is exponential with mean $1/\lambda$. Hence, this is just the Poisson process.

Ex 4.2.3 (M/M/1 Queueing System) Suppose that customers arrive at a single-server service station with a Poisson process having rate λ. Upon arrival, each customer goes directly into service if the server is free; if not, then the customer joins the queue. When the server finishes serving a customer, the customer leaves the system and the next customer in line, if there are any waiting, enters the service. This is known as M/M/1 queueing system.

If we let $X(t)$ denote the number of customers in the system at time t then $\{X(t), t \geq 0\}$ is a birth and death process with

$$\mu_n = \mu, \qquad n \geq 1$$

$$\lambda_n = \lambda, \qquad n \geq 0$$

Ex 4.2.4 (A multiserver exponential queueing system) Consider an exponential queueing system in which there are s servers available, each server at rate μ. An entering customer first waits in line and then goes to the first free server. This is a birth and death process with parameters

$$\mu_n = \begin{cases} n\mu, & 1 \leq n \leq s \\ s\mu, & n > s \end{cases}$$

Balance Equation for Birth and Death process

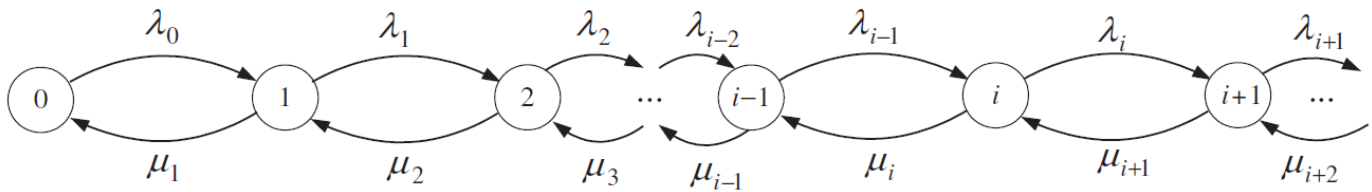


Figure 5.3 State-transition-rate diagram for birth and death process.

In the steady state, $p_j(t) \to p_j$ and

$$\lim_{t \to \infty} \left\{ \frac{\mathrm{d}p_i(t)}{\mathrm{d}t} \right\} = 0$$

Rate at which enters tate i

Rate at which leaves state i

Balance Equation for the birth and death process

$$\lambda_0 p_0 = \mu_1 p_1$$

$$(\lambda_i + \mu_i) p_i = \mu_{i+1} p_{i+1} + \lambda_{i-1} p_{i-1} \qquad i = 1, 2, \ldots$$

$$\sum_i p_i = 1$$

Transition from state i

Birth or death occurs first

Limiting probabilities

In analogy with DTMC, the probability that a CTMC will be in state j at time t often converges to a limiting value which is independent of the initial state:

$$P_i = \lim_{t \to \infty} P_i(t)$$

Solution of the system of equations

A sufficient condition for existence of limiting probabilities P_j is that

i. All states of the MC communicate.

ii. The MC is positive recurrent.

In steady state the rate at which a process enter a state is equal to the rate at which the process leaves the state

States Rate at which leave = Rate at which enter

0

1

2

….. ……..

…..n

By adding to each equation the equation preceding it

And by using the fact that $\sum_{n=0}^{\infty} P_n = 1$, we obtain

$$1 = P_0 + P_0 \sum_{n=1}^{\infty} \frac{\lambda_{n-1}\lambda_{n-2}\cdots\lambda_1\lambda_0}{\mu_n\mu_{n-1}\cdots\mu_2\mu_1}$$

$$P_0 = \frac{1}{1 + \sum_{n=1}^{\infty} \frac{\lambda_{n-1}\lambda_{n-2}\cdots\lambda_1\lambda_0}{\mu_n\mu_{n-1}\cdots\mu_2\mu_1}}$$

$$P_n = \frac{\lambda_{n-1}\lambda_{n-2}\cdots\lambda_1\lambda_0}{\mu_n\mu_{n-1}\cdots\mu_2\mu_1\left(1 + \sum_{n=1}^{\infty} \frac{\lambda_{n-1}\lambda_{n-2}\cdots\lambda_1\lambda_0}{\mu_n\mu_{n-1}\cdots\mu_2\mu_1}\right)}, \quad n \geq 1$$

From the above, the necessary condition for the limiting probabilities to exist is

$$\sum_{n=1}^{\infty} \frac{\lambda_{n-1}\lambda_{n-2}\cdots\lambda_1\lambda_0}{\mu_n\mu_{n-1}\cdots\mu_2\mu_1} < \infty$$

This condition also may be shown to be sufficient.

Ex 4.4.2 (Ex 6.15, pp. 374) Consider the showshine shop and determine the proportion of the time the process is in each of the states 0, 1, 2. Note that it is not a birth and death process.

$$P_1 = \frac{\lambda\mu_2}{\mu_1\mu_2 + \lambda(\mu_1 + \mu_2)}$$

$$P_2 = \frac{\lambda\mu_1}{\mu_1\mu_2 + \lambda(\mu_1 + \mu_2)}$$

Consider a job shop that consists of M machines and one serviceman. Suppose that the amount of time each machine runs before breaking down is exponentially distributed with mean $\frac{1}{\lambda}$, and suppose that the amount of time that it takes for the serviceman to fix a machine is exponentially distributed with mean $\frac{1}{\mu}$. We shall attempt to answer these questions: (a) What is the average number of machines not in use? (b) What proportion of time is each machine in use?

$$\frac{1}{1 + \sum_{n=1}^{M} (\lambda/\mu)^n M!/(M-n)!}$$

Example 5.4

Consider the Markov chain whose state-transition-rate diagram is given in Find m_{14}.

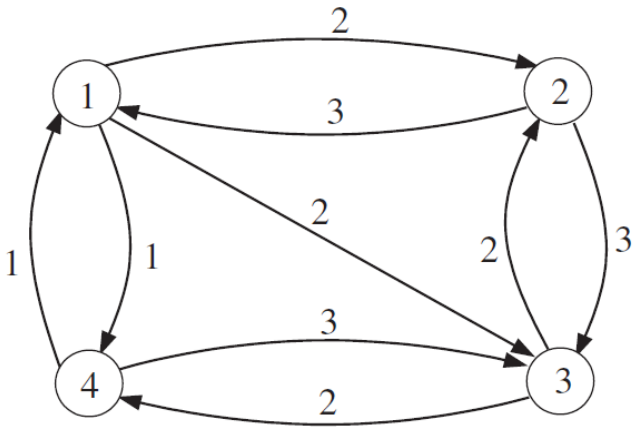


Figure 5.5 State-transition-rate diagram for Example 5.4.